A Short Note on Heavy-Tail Combination Tests

Yikun Zhang

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This note is mainly based on Professor Xiao-Li Meng's talk "*From A Cauchy Surprise to the Half-Cauchy Miracle*" at the Statistics seminar on October 23, 2026 at Northwestern University.[1] Since there is no recording, I wrote this note based on my own understanding and wording. All insights in this note should be credited to Professor Meng's enlightening talk, while any errors are entirely my own.

1 A Cauchy Surprise

The nature world can be both good and evil, and so can the Cauchy distribution. On the one hand, to quote Pillai and Meng (2016), "some introductory courses have given the Cauchy distribution the nickname 'Evil', because it has created a few excruciating moments..." The major reason is that the Cauchy distribution has no mean (*i.e.*, its expectation is undefined) due to the heavy-tailed characteristics. It is intuitive (or counterintuitive) to see from the probability density function (PDF) of a general Cauchy distribution Cauchy(μ, σ):

$$f(x;\mu,\sigma) = \frac{1}{\pi\sigma\left[1+\left(\frac{x-\mu}{\sigma}\right)^2\right]}, \qquad x \in \mathbb{R}, \tag{1}$$

where $\mu \in \mathbb{R}$ is the location parameter for the mode of the PDF, while σ is the scale parameter that specifies the half-width at half-maximum. For the standard Cauchy distribution, $\mu = 0$ and $\sigma = 1$. The mean of the standard Cauchy distribution, if it exists, would be written as

$$\int_{\mathbb{R}} x \cdot f(x;0,1)\,dx = \int_a^{\infty} x \cdot f(x;0,1)\,dx + \int_{-\infty}^{a} x \cdot f(x;0,1)\,dx \qquad \text{for any } \ a \in \mathbb{R}.$$

Since both integrals on the right-hand side are infinite and of the opposite sign, the integral on the left-hand side does not exist. Furthermore, the Cauchy distribution does not have any finite moment of any order $k \geq 1$, though the even-powered moment $\mathbb{E}(X^k)$ does exist (and equal to ∞).

On the other hand, the Cauchy distribution has a "nice" property that the sample mean $\bar{X}_n = \frac{1}{n}\sum_{i=1}^n X_i$ of independent and identically distributed (i.i.d.) standard Cauchy random variables $X_1, ..., X_n$ preserves its distribution, *i.e.*, $\bar{X}_n \sim$ Cauchy$(0,1)$, however large n goes, where the symbol "$\sim$ means "is distributed like". Indeed, the Cauchy distribution is an example of the more general stable distributions, where for three *independent* copies X_1, X_2, X_3 from a stable distribution, $aX_1 + bX_2$ has the same distribution as $cX_3 + d$ for some constants $a, b, c > 0$ and

[1] I have long heard (from YouTube videos) that Professor Meng is an extraordinary speaker, so I was not surprised to see a large turnout for his talk, especially considering that Northwestern's Statistics Department is relatively small and the talk was not widely advertised. What surprised me, though, was that no one opened their cellphone and did anything unrelated to his talk, even though at least half the attendees were students required to attend the seminar. His talk really made my 2-hour round trip to Evanston well worth it and inspired both how I would give my talk in the following week and this note.

$d \in \mathbb{R}$. As a side note, this property also leads to the Cauchy version of the central limit theorem (CLT):

Theorem 1.1: Cauchy CLT. If $X_1, X_2, ...$ are i.i.d. observations from a distribution with density p such that $\lim_{c\to\infty} \frac{1}{c}\int_{-c}^{c} x^2 p(x)\,dx = \frac{2\gamma}{\pi} \in (0, \infty)$, then

$$\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i \xrightarrow{d} \text{Cauchy}(\mu, \gamma) \quad \text{for some constant } \mu \in \mathbb{R}.$$

What is surprising is the following theorem. It is well-known (from direct calculations or polar coordinate transformation) that, if X and Y are independent Gaussian random variables from $\mathcal{N}(0, \sigma^2)$, then both $\frac{X}{Y}$ and $\frac{Y}{X}$ have the standard Cauchy distribution Cauchy$(0, 1)$. However, Drton and Xiao (2016)[2] conjectured the following result, which was then proved by Pillai and Meng (2016).

Theorem 1.2. Let $\boldsymbol{X} = (X_1, ..., X_d)$ and $\boldsymbol{Y} = (Y_1, ..., Y_d)$ be i.i.d. $\mathcal{N}(\boldsymbol{0}, \Sigma)$, where $\Sigma = \{\sigma_{ij}\} \geq 0$ is an *arbitrary* covariance matrix in $\mathbb{R}^{d\times d}$ with $\sigma_{ii} > 0$ for all $1 \leq i \leq d$. If $\boldsymbol{w} = (w_1, ..., w_d)$ is a (random) vector independent of $\boldsymbol{X}$ and $\boldsymbol{Y}$ with $\sum_{j=1}^{d} w_j = 1$ and $w_j \geq 0$ for $j = 1, ..., d$, then

$$Z = \sum_{j=1}^{d} w_j \cdot \frac{X_j}{Y_j} \sim \text{Cauchy}(0, 1). \tag{2}$$

There are two remarkable points in Theorem 1.2. First, Σ is not necessarily positive definite and Σ^{-1} may not exist. Second and more surprisingly, those terms $w_j \cdot \frac{X_j}{Y_j}$ in (2) are no longer independent. Nevertheless, Z is still Cauchy-distributed despite the fact that we take a convex combination of dependent Cauchy random variables. How unexpected it is when we compare this result with the usual setting in Cauchy CLT (Theorem 1.1).

To add an extra layer of surprise, Pillai and Meng (2016) presented a astonishingly simple proof of Theorem 1.2 based on the following lemma in Pitman and Williams (1967); Williams (1969), though the proof of this lemma itself requires techniques from the complex analysis.[3]

Lemma 1.3. Let $\Theta \sim \text{Uniform}\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, and $\{w_1, ..., w_d\}$ independent of Θ, where $w_j \geq 0$ and $\sum_{j=1}^{d} w_j = 1$. Then, for any $\{u_1, ..., u_d\}$ with $u_j \in \mathbb{R}$,

$$\sum_{j=1}^{d} w_j \tan(\Theta + u_j) \sim \text{Cauchy}(0, 1).$$

In fact, Williams (1969) proved a quasi-reverse of Lemma 1.3: if u_j is not a rational multiple of π and $\tan(\Theta + u_j) \sim \tan\Theta$, then $X = \tan\Theta$ has to be Cauchy$(0, 1)$.

With Lemma 1.3, Pillai and Meng (2016) proved Theorem 1.2 by using the polar coordinate

[2] I started my graduate study at the Department of Statistics at the University of Washington after Professor Mathias Drton left the department, but I was fortunate to chat with Professor Drton 1:1 during his homecoming seminar in October 2023. I was very impressed by the strengths of Professor Drton, both mathematically based on his talk and physically based on our handshake.

[3] It also reminds me of another Cauchy-distributed result for Brownian motions that I proved 7 years ago.

transformation $X_j = R_j \cos\Theta_j, Y_j = R_j \sin\Theta_j$ for $j = 1, ..., d$ and a further transformation $U_j = (\Theta_j - \Theta_1) \mod (2\pi)$ for $j = 2, ..., d$ as well as noticing that $U_j, j = 2, ..., d$ are independent of $\Theta_1 \sim \text{Uniform}\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ and

$$Z = \sum_{j=1}^{d} w_j \cdot \frac{X_j}{Y_j} = \sum_{j=1}^{d} w_j \cdot \tan\Theta_j = \sum_{j=1}^{d} w_j \cdot \tan(\Theta_1 + U_j).$$

2 A Cauchy Blessing for Multiple Testing

One immediate application of Theorem 1.2 is the Wald test for a homogeneous d-variate polynomial $q(x_1, ..., x_d) = x_1^{a_1} \cdots x_d^{a_d}$ with gradient ∇q, $a_j \geq 0$, and $\sum_{j=1}^{d} a_j > 0$. By the delta method and $\boldsymbol{X} \sim \mathcal{N}(\mathbf{0}, \Sigma)$,

$$\begin{aligned}
W_{q,\Sigma} := \frac{q(\boldsymbol{X})^2}{[\nabla q(\boldsymbol{X})]^T \Sigma \nabla q(\boldsymbol{X})} &\equiv \left\{ [\nabla \log |q(\boldsymbol{X})|]^T \Sigma [\nabla \log |q(\boldsymbol{X})|] \right\}^{-1} \sim \frac{1}{\left(\sum_{j=1}^{d} a_j\right)^2} \chi_1^2, \\
\iff & \left(\frac{a_1}{X_1}, ..., \frac{a_d}{X_d}\right)^T \Sigma \left(\frac{a_1}{X_1}, ..., \frac{a_d}{X_d}\right) \sim \chi_1^{-2}, \\
\iff & \text{let } Q := \frac{[\nabla \log |q(\boldsymbol{X})|]^T \boldsymbol{Y}}{\sqrt{[\nabla \log |q(\boldsymbol{X})|]^T \Sigma \nabla \log |q(\boldsymbol{X})|}} \quad \text{and } \boldsymbol{Y} \sim \mathcal{N}(\mathbf{0}, \Sigma), \quad \boldsymbol{Y} \perp\!\!\!\perp \boldsymbol{X} \text{ so that} \\
& Q|\boldsymbol{X} \sim \mathcal{N}(0, 1),\ Q \perp\!\!\!\perp W_{q,\Sigma} \text{ and } \frac{Q}{\sqrt{W_{q,\Sigma}}} = \sum_{j=1}^{d} a_j \cdot \frac{Y_j}{X_j} = A \sum_{j=1}^{d} \frac{a_j}{A} \cdot \frac{Y_j}{X_j} \sim \text{Cauchy}(0, A) \\
& \text{with } A = \sum_{j=1}^{d} a_j,
\end{aligned}$$

where χ_1^2 denotes a standard chi-squared random variable with 1 degree of freedom, and χ_1^{-2} is an inverse chi-squared random variable with 1 degree of freedom.

A more interesting application lies in multiple testing problems, where individual p-values or test statistics from several studies are combined to simultaneously assess the statistically significant effects. Often, the aggregation of these p-values requires independent or weakly dependent structures among them, or distributional assumptions on test statistics under the null hypotheses. One of the most famous examples is the Fisher's combination test: given k *independent* p-values $p_1, ..., p_k$ from separate studies, we transform them using the natural logarithm and derive the test statistic as

$$T_{\text{Fisher}} = -2 \sum_{i=1}^{k} \log p_i. \tag{3}$$

Under the global null hypothesis $H_0 = \cap_{i=1}^{k} H_{0,i}$ (*i.e.*, all the null hypotheses $H_{0,i}, i = 1, ..., k$ are true) and independence between $p_i, i = 1, ..., k$, $T_{\text{Fisher}} \sim \chi_{2k}^2$. Another way to combine *independent* p-values is to take the minimum $p_{(1)} = \min\{p_1, ..., p_k\}$, called the Tippett's minimum p-value, whose reference distribution under the null is a Beta distribution $\text{Beta}(1, k)$.

In practice, however, the p-values (or test statistics) across studies could be highly dependent. When dependence among $p_1, ..., p_k$ is positive, the Fisher's combination test statistic T_{Fisher} is overly

large, or equivalently, the p-value of T_{Fisher} is too small based on the above χ^2_{2k} distribution under the global null, leading to an anti-conservative significant conclusion on the aggregated effect.

Interestingly, the heavy-tailed characteristic of the Cauchy distribution and Theorem 1.2 offer a solution to combining dependent p-values for multiple testing (Liu and Xie, 2020). Given the individual p-values $p_i, i = 1, ..., k$, the Cauchy combination test (CCT) has the following test statistic as

$$T_{\text{Cauchy}} = \sum_{i=1}^{k} w_i \tan\left[(0.5 - p_i)\pi\right] = \sum_{i=1}^{k} w_i \cot(p_i\pi), \tag{4}$$

where the weights satisfy $w_i \geq 0$ and $\sum_{i=1}^{k} w_i = 1$. By Lemma 1.3, when p_i's are independent or perfectly dependent, T_{Cauchy} has a standard Cauchy distribution under the global null. When p_i's (or their corresponding mean-zero test statistics X_i's) are correlated but not perfectly dependent, Liu and Xie (2020) shows that

$$\lim_{t\to\infty} \frac{\mathbb{P}\left(T_{\text{Cauchy}} > t\right)}{\mathbb{P}\left(\text{Cauchy}(0,1) > t\right)} = 1$$

under a fixed number of studies k or a diverging number of studies $k = o(t^c)$ for some constant $c \in (0, 0.5)$ and a pairwise bivariate normal assumption on X_i's. The bivariate normal assumption was later relaxed by Long et al. (2023), and they incorporated several bivariate copulas as examples. As a side note, another statistical technique for combining dependent p-values is the harmonic mean p-value (Wilson, 2019):

$$\mathring{p} = \frac{\sum_{i=1}^{k} w_i}{\sum_{i=1}^{k} \frac{w_i}{p_i}}, \tag{5}$$

where the weights again satisfy $w_i \geq 0$ and $\sum_{i=1}^{k} w_i = 1$. Its asymptotically exact p-value under the global null and large k can be computed as

$$p_{\mathring{p}} = \int_{1/\mathring{p}}^{\infty} f_{\text{Landau}}\left(x \middle| \log k + 0.874, \frac{\pi}{2}\right) dx \quad \text{with} \quad f_{\text{Landau}}(x \mid \mu, \sigma) = \frac{1}{\pi}\int_0^{\infty} e^{-\frac{(x-\mu)t}{\sigma} - \frac{2t\log t}{\pi}} \sin(2t)\, dt;$$

see also Goeman et al. (2019) for discussion on, whether or to which level, the harmonic mean p-value control familywise error rate (FWER).

3 A Half-Cauchy Improvement

One drawback of the CCT is that T_{Cauchy} in (4) approaches $-\infty$ when one or several p-values approach 1. This property is less ideal because among studies to be tested simultaneously, only a small portion of the p-values are small, while other p-values could be largely close to 1. A single "highly insignificant" test could substantially degrade the power of the combined test. Furthermore, based on the duality between hypothesis testing and confidence regions, the resulting confidence region from the CCT is not necessarily convex (or not a single interval for univariate cases; Meng 2024). This non-convex feature also makes the construction of confidence regions from CCT difficult.

Specifically, the asymptotic $(1-\alpha)$-level confidence interval for $\theta \in \mathbb{R}$ or confidence region for

$\boldsymbol{\theta} \in \mathbb{R}^d$ as $\alpha \to 1$ is given by

$$\left\{\theta \in \mathbb{R} : \sum_{j=1}^{k} w_j \cot\left\{2\pi\left[1 - \Phi_j\left(\frac{|\widehat{\theta}_j - \theta|}{\widehat{\sigma}_i}\right)\right]\right\} \leq \cot(\pi\alpha)\right\}, \tag{6}$$

$$\left\{\boldsymbol{\theta} \in \mathbb{R}^d : \sum_{j=1}^{k} w_j \cot\left[2\pi\left\{1 - F_j\left(\left(\widehat{\boldsymbol{\xi}}_j - P_j\boldsymbol{\theta}\right)^T \widehat{\Sigma}_j^{-1}\left(\widehat{\boldsymbol{\xi}}_j - P_j\boldsymbol{\theta}\right)\right)\right\}\right] \leq \cot(\pi\alpha)\right\}, \tag{7}$$

where $\widehat{\theta}_j$ and $\widehat{\sigma}_j$ are estimators of θ and the *standard error* of the population from the j-th study, respectively, and Φ_j is the cumulative distribution function (CDF) of the t-distribution with m_j degrees of freedom, which could be $m_j = \infty$ for the normal CDF. Additionally, $P_j \in \mathbb{R}^{d_j \times d}$ is a (user-chosen) full-rank matrix with $d_j \leq d$, $\widehat{\boldsymbol{\xi}}_j \in \mathbb{R}^{d_j}$ is an estimator from the j-th study for $P_j\boldsymbol{\theta}$, and F_j is the CDF of the Hotelling's T^2-distribution $T^2(d_j, k_j)$ with k_j degrees of freedom with $\widehat{\Sigma}_j$, which has the same distribution as the scaled F-distribution $\frac{d_j k_j}{k_j + 1 - d_j} F(d_j, k_j + 1 - d_j)$. For $k = 1$, (6) reduces to the usual two-sided t-test confidence interval $\left[\widehat{\theta}_1 \pm t_{1-\alpha/2,k_1}\widehat{\sigma}_1\right]$. For (7), we always choose P_j so that $\widehat{\Sigma}_j$ can be constructed as a positive definite covariance estimator of $\widehat{\boldsymbol{\xi}}_j$. In practice, it is recommended to choose $d_j = 1$.

Now, let $p_j(\theta) = 2\left[1 - \Phi_j\left(\frac{|\widehat{\theta}_j - \theta|}{\widehat{\sigma}_i}\right)\right]$ in (6). It is easy to see that for any $w_j > 0$, $\widehat{\theta}_j$ always lies in (6), because $p_j(\widehat{\theta}_j) = 1$ and $\cot\left[\pi p_j(\widehat{\theta}_j)\right] = -\infty$. Thus, (6) may have several disconnected components centered at $\widehat{\theta}_j, j = 1, ..., k$.

To address the above instability and non-convex confidence region issues of the CCT, Liu et al. (2025) proposed a clever half-Cauchy combination test (HCCT) with test statistic as

$$T_{\text{HCCT}} = \sum_{i=1}^{k} w_j \cdot \cot\left(\frac{\pi p_j}{2}\right), \tag{8}$$

where the weights again satisfy $w_i \geq 0$ and $\sum_{i=1}^{k} w_i = 1$. Different from (4), any study with $p_j = 1$ will contribute 0 instead of $-\infty$ to the test statistic, and $T_{\text{HCCT}} \geq 0$. Moreover, given that $p_j(\theta) = 2\left[1 - \Phi_j\left(\frac{|\widehat{\theta}_j - \theta|}{\widehat{\sigma}_i}\right)\right]$,

$$\begin{aligned}
T_{\text{HCCT}} &= \sum_{j=1}^{k} w_j \cdot \cot\left[\pi - \pi \cdot \Phi_j\left(\frac{|\widehat{\theta}_j - \theta|}{\widehat{\sigma}_i}\right)\right] \\
&= -\sum_{j=1}^{d} w_j \cdot \cot\left[\pi \cdot \Phi_j\left(\frac{|\widehat{\theta}_j - \theta|}{\widehat{\sigma}_i}\right)\right] \\
&\overset{\text{(i)}}{\geq} -\sum_{j=1}^{d} w_j \cdot \cot\left[\arctan\left(\frac{|\widehat{\theta}_j - \theta|}{\widehat{\sigma}_i}\right) + \frac{\pi}{2}\right] \\
&= \sum_{j=1}^{d} w_j \cdot \frac{|\widehat{\theta}_j - \theta|}{\widehat{\sigma}_i} \\
&\geq \sum_{j=1}^{d} w_j \cdot \frac{|\widehat{\theta}_j - \widehat{\theta}^*|}{\widehat{\sigma}_i}
\end{aligned}$$

where (i) follows from the fact that $\Phi_j(x) \geq F_{\text{Cauchy}}(x) = \frac{\arctan(x)}{\pi} + \frac{1}{2}$ for either t or normal CDF Φ_j, and $\widehat{\theta}^*$ is the minimizer of the weighted L_1-loss $\sum_{j=1}^d w_j \cdot \frac{|\widehat{\theta}_j - \theta|}{\widehat{\sigma}_i}$. In other words, $\{\theta \in \mathbb{R} : T_{\text{HCCT}} \leq K\}$ can be empty when, *e.g.*, $K \leq \sum_{j=1}^d w_j \cdot \frac{|\widehat{\theta}_j - \widehat{\theta}^*|}{\widehat{\sigma}_i}$. Liu et al. (2025) further showed that the confidence region

$$\left\{ \boldsymbol{\theta} \in \mathbb{R}^d : \sum_{j=1}^k w_j \cot\left[\pi\left\{1 - F_j\left(\left(\widehat{\boldsymbol{\xi}}_j - P_j\boldsymbol{\theta}\right)^T \widehat{\Sigma}_j^{-1}\left(\widehat{\boldsymbol{\xi}}_j - P_j\boldsymbol{\theta}\right)\right)\right\}\right] \leq \cot\left(\frac{\pi\alpha}{2}\right)\right\}, \tag{9}$$

from HCCT is convex when $d_j = 1, k_j \geq 1$ or $k_j \geq d_j + 1$.

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